

Research Questions

Defending Tracking Control Neural Networks
Against Adversarial Attacks on ACC Systems

Ablasse Kingcaid-Ouedraogo

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1 How Does My Paper Relate to the ACC/IDS Problem?

Background: The Chain of Papers

The existing research establishes:

1. **Adaptive Cruise Control (ACC)** regulates vehicle speed and following distance using a PID controller. The PID computes a control signal from the speed error:

$$u(t) = K_p e(t) + K_i \int_0^t e(\tau) d\tau + K_d \frac{de(t)}{dt}$$

where $e(t) = S_t(t) - S_c(t)$ is the difference between target speed and current (sensor-reported) speed.

2. **The Intrusion Detection System (IDS)** is a passive extension to the ACC. It monitors the **Control Area Network (CAN)** bus for spoofed speed messages using machine learning. When it detects an attack, it raises a flag.
3. **Paper [1]** investigates the impact of **cold braking** as a mitigation strategy.

Question: What does my paper do differently?

My paper investigates an alternative mitigation approach. Instead of cold braking when the IDS detects an attack, I propose:

- (a) Using the **tracking control equation**_[2] to continuously track the error between the expected speed (what the vehicle should be doing based on the reference trajectory) and the reported speed (what the potentially-spoofed sensor claims).
- (b) **Correcting for the error** using the designed control function $u(k)$ _[2], which computes the input needed to drive the system state back toward the reference trajectory.
- (c) Performing a **mode switch** if the tracking error exceeds a threshold, i.e. if the discrepancy between expected and reported speed grows too large, the system can escalate its response (e.g., increase correction aggressiveness or fall back to emergency braking).

Additionally, I **simulate the attack scenarios** to demonstrate that the tracking controller successfully corrects for injected speed errors, showing convergence of the tracking error to zero (or near-zero) even under adversarial perturbation.

Summary of Positioning

Component	Paper [1] Approach	My Approach
Detection	ML-based IDS on CAN bus	Same (IDS flags the attack)
Mitigation	Cold brake (binary)	Tracking control correction
Error model	PID speed error $e(t)$	Discrete-time $e(k) = r(k) - x(k)$
Correction	None (just stop)	$u(k)$ drives $e(k) \rightarrow 0$
Escalation	N/A	Mode switch if $\ e(k)\ > \theta$

References

- [1] M. B. Jedh, L. ben Othmane, and A. K. Somani, “Improvement and evaluation of resilience of adaptive cruise control against spoofing attacks using intrusion detection system,” 2023. arXiv:2302.00876v1.
- [2] P. Tymoshchuk, “A tracking control neural network for the known affine in the inputs discrete-time nonlinear systems,” 2024. University of North Texas.

2 Is My Variable Mapping to the ACC Domain Correct?

The Tracking Control System Model

The system model from the paper is:

$$x(k+1) = x(k) + \tau [f(x(k)) + g(x(k)) u(k)] \quad (\text{System Model})$$

With the designed control function, the deployed tracking equation becomes:

$$x(k+1) = x(k) - \beta \operatorname{sgn}(x(k) - r(k)) + \Delta r(k) \quad (\text{Tracking Eq.})$$

where $\Delta r(k) = r(k+1) - r(k)$ and the error dynamics satisfy:

$$e(k+1) = e(k) - \beta \operatorname{sgn}(e(k)) \quad (\text{Error Dynamics})$$

Question: Does this variable mapping hold?

Variable	General Meaning	ACC Interpretation
$r(k)$	Reference trajectory	Expected speed at time step k
$\Delta r(k)$	Change in reference	Expected change in speed
$x(k)$	System state	Reported (sensor) speed at time step k
$e(k)$	Tracking error $r(k) - x(k)$	Discrepancy: expected vs. reported speed
$u(k)$	Control input	Corrective action applied to reduce error
$g(x(k))$	Input coupling function	How the correction maps to state change

Analysis: What Needs Clarification

The mapping is **mostly correct**, but there are nuances to think through:

1. $r(k)$ — **“Expected speed”**: This is the reference trajectory that the vehicle *should* follow. In the ACC context, this comes from the upper controller’s target speed profile (based on the driver’s set speed and the safe following distance). This is well-defined.
2. $x(k)$ — **“Reported speed”**: Here is where the attack matters. Under normal operation, $x(k)$ is the true vehicle speed. Under a spoofing attack, $x(k)$ becomes the *corrupted* sensor reading. The question is: does the tracking controller operate on the reported (possibly spoofed) value, or does it have access to a separate estimate of the true speed?
 - If the controller only sees the spoofed $x(k)$, then $e(k) = r(k) - x(k)$ is *itself corrupted*, and the controller will issue wrong corrections.
 - If the controller has an independent estimate of true speed (e.g., from wheel encoders, IMU integration, or the IDS’s detection), then $x(k)$ represents this estimate, and the tracking controller corrects against the attack.

This distinction is critical to define clearly in the paper.

3. $u(k)$ — **“Correction function”**: In the general system model, $u(k)$ is the full control function:

$$u(k) = g(x(k))^{-1} (-f(x(k)) + \Delta r(k) - \beta \operatorname{sgn}(x(k) - r(k)))$$

In the ACC analog, this represents the acceleration/deceleration command that the tracking controller issues to correct the speed error. This is analogous to (but distinct from) the PID's $u(t)$.

4. $g(x(k))$ — **“How correction affects reported speed”**: In the general model, $g(x(k))$ is the input coupling — it determines how effectively the control input influences the state dynamics. In the ACC context, this captures the vehicle's response characteristics (e.g., how throttle/brake commands translate to speed changes, which depends on current speed, road grade, etc.).

3 How Will Simulated Attacks Be Represented in the Math?

Question

The tracking control equations define normal system behavior. How do we mathematically represent an adversarial attack specifically, CAN bus speed spoofing, within this framework?

Is this an open research question?

4 Where Is the Attack Function in Paper [3]?

Question

Paper [3] (Jedh et al.) simulates speed spoofing attacks on the ACC. Where in the paper is the mathematical representation of the simulated attacks, and how are they parameterized?

Answer: Paper [3] Does Not Define a Formal Attack Function

Paper [3] does **not** provide a closed-form mathematical function for the attack signal $a(k)$. Instead, the attacks are defined **operationally** through the CARLA simulation setup. Specifically:

1. **Section 3.2** (Architecture of the Extended Adaptive Cruise Control) describes the attack conceptually: spoofed speed messages are injected into the CAN bus, and the IDS monitors for these injections.
2. **Section 4.1** (Simulation Setup) defines the attack parameters for each scenario:
 - **Scenario 2:** The ego vehicle sensor speed is spoofed to 10 km/h (a constant offset). The attack interval is fixed at a rate of 75%, meaning spoofed messages replace real ones with probability $p = 0.75$. No IDS is used.
 - **Scenario 3:** Same spoofing (10 km/h), same injection probability ($p = 0.75$), but now the ACC is extended with the real-time IDS. When the IDS detects the spoofing, it triggers cold braking.
3. **Section 4.2** (Simulation Results) shows the effects in plots but does not formalize the injection mathematically.

Reconstructing the Attack Model from Paper [3]

From the simulation parameters, the implicit attack model is:

$$\tilde{x}(k) = \begin{cases} x_{\text{spoof}} & \text{with probability } p \\ x(k) & \text{with probability } 1 - p \end{cases} \quad (\text{Paper [3] Attack Model})$$

where:

x_{spoof} The constant spoofed speed value (10 km/h in Paper [3]).

p Injection probability (0.75 in Paper [3]).

$x(k)$ True vehicle speed at time step k .

$\tilde{x}(k)$ Speed value actually received by the ACC controller.

This can equivalently be written using an indicator random variable $B(k) \sim \text{Bernoulli}(p)$:

$$\tilde{x}(k) = (1 - B(k)) x(k) + B(k) x_{\text{spoof}} \quad (\text{Equivalent Form})$$

Or as an additive attack signal:

$$a(k) = B(k)(x_{\text{spoof}} - x(k)) \quad \text{so that} \quad \tilde{x}(k) = x(k) + a(k) \quad (\text{Additive Form})$$

Note that this attack signal $a(k)$ is **state-dependent** — its magnitude depends on the current true speed $x(k)$, since it replaces the true value with a fixed spoofed value rather than adding a fixed offset.

Implication for Your Paper

Since Paper [3] defines attacks only through simulation parameters (not formal equations), your paper has the opportunity to:

- (a) Provide the **formal mathematical attack model** that Paper [3] lacks.
- (b) Show how this attack model integrates into the tracking control framework's error dynamics.
- (c) Demonstrate via simulation that the tracking controller handles the attack, which Paper [3] addresses only through the binary cold-brake response.

This is a clear contribution: formalizing what Paper [3] left implicit, and providing a more nuanced mitigation than emergency braking.